

End-to-End Derivation: From Maxwell's Equations to the FEM Linear System Radar Scattering from Heterogeneous Rough Soil

Abstract

This document derives, step by step and without skipping algebra, the full chain from the time-harmonic Maxwell equations to the assembled finite-element linear system $A\mathbf{u} = \mathbf{b}$ used to compute electromagnetic scattering from a rough, depth-heterogeneous soil surface. Each stage states *what* is being done and *why*, so the derivation can be followed independently of the source paper.

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1 From Maxwell's Equations to the Vector Wave Equation

1.1 Starting point: time-harmonic Maxwell's equations

Assume a time-harmonic convention $e^{j\omega t}$ and a source-free region. Maxwell's curl equations are

$$\nabla \times \mathbf{E} = -j\omega\mu_0\mu_r\mathbf{H}, \quad (1)$$

$$\nabla \times \mathbf{H} = j\omega\varepsilon_0\varepsilon_r\mathbf{E}, \quad (2)$$

where $\varepsilon_r = \varepsilon_r(\mathbf{r})$ may vary with position (this is exactly how soil moisture enters the problem, via $\varepsilon_r = \varepsilon_r(m_v(z))$), while the soil is assumed non-magnetic, $\mu_r = 1$, constant everywhere.

Define the free-space wavenumber

$$k_0^2 = \omega^2\mu_0\varepsilon_0.$$

1.2 Eliminating one field: two possible reductions

We want a single second-order PDE in *one* unknown field. There are two ways to do this, and which one is convenient depends on which field's governing equation has the position-dependent coefficient trapped *inside* a curl versus multiplying *outside* it.

Reduction A: solving for $\mathbf{E}$ (used for TE polarization). Take the curl of (1). Since $\mu_r = 1$ is constant, it can be pulled outside:

$$\nabla \times (\nabla \times \mathbf{E}) = -j\omega\mu_0 \nabla \times \mathbf{H}.$$

Substitute (2) for $\nabla \times \mathbf{H}$:

$$\nabla \times (\nabla \times \mathbf{E}) = -j\omega\mu_0(j\omega\varepsilon_0\varepsilon_r\mathbf{E}) = \omega^2\mu_0\varepsilon_0\varepsilon_r\mathbf{E} = k_0^2\varepsilon_r\mathbf{E}.$$

Hence

$$\nabla \times (\nabla \times \mathbf{E}) - k_0^2\varepsilon_r\mathbf{E} = 0. \quad (3)$$

Here ε_r multiplies $\mathbf{E}$ *outside* any curl — no obstruction from its spatial dependence.

Reduction B: solving for $\mathbf{H}$ (used for TM polarization). This time we cannot immediately take the curl of (2) and pull ε_r out, because ε_r varies with position: $\nabla \times (\varepsilon_r\mathbf{E}) \neq \varepsilon_r \nabla \times \mathbf{E}$ in general. Instead, solve (2) for $\mathbf{E}$,

$$\mathbf{E} = \frac{1}{j\omega\varepsilon_0\varepsilon_r} \nabla \times \mathbf{H},$$

and substitute into (1):

$$\nabla \times \left(\frac{1}{j\omega\varepsilon_0\varepsilon_r} \nabla \times \mathbf{H} \right) = -j\omega\mu_0\mathbf{H} \implies \frac{1}{j\omega\varepsilon_0} \nabla \times \left(\frac{1}{\varepsilon_r} \nabla \times \mathbf{H} \right) = -j\omega\mu_0\mathbf{H}.$$

Multiplying both sides by $j\omega\varepsilon_0$:

$$\nabla \times \left(\frac{1}{\varepsilon_r} \nabla \times \mathbf{H} \right) - k_0^2 \mathbf{H} = 0. \quad (4)$$

Here $1/\varepsilon_r$ is trapped *inside* the first curl, because that is the only way to legally move the spatially-varying coefficient across the derivative.

1.3 Unified form

Equations (3) and (4) are both special cases of a single template equation for a field U :

$$\boxed{\nabla \times \left(\frac{1}{p} \nabla \times U \right) - k_0^2 q U = 0} \quad (5)$$

with

$$\text{TM: } U = H_y, \quad p = \varepsilon_r, \quad q = \mu_r, \quad \text{TE: } U = E_y, \quad p = \mu_r, \quad q = \varepsilon_r,$$

matching (4) ($p = \varepsilon_r, q = \mu_r = 1$) and (3) ($p = \mu_r = 1, q = \varepsilon_r$) respectively. In the 2-D problem ($\partial/\partial y \equiv 0$), U is the single nonzero out-of-plane field component, and the double curl in (5) reduces to a 2-D differential operator acting on that scalar-valued component.

Why this matters: the choice of which field to solve for is not arbitrary — it is dictated by needing the spatially-varying material coefficient to sit *inside* the outer curl, so that the equation remains valid even when $\varepsilon_r = \varepsilon_r(z)$ is depth-dependent (the heterogeneous-soil case).

2 Weak (Variational) Formulation

2.1 The residual and the weighted-residual statement

Let U^h be any approximation to the exact field U . Substituting U^h into (5) generally leaves a nonzero residual,

$$R(\mathbf{r}) = \nabla \times \left(\frac{1}{p} \nabla \times U^h \right) - k_0^2 q U^h \neq 0.$$

Rather than demanding $R(\mathbf{r}) = 0$ pointwise (which would require U^h to be exact), FEM demands that R vanish in a *weighted-average* sense against an arbitrary test function T :

$$\iint_{\Omega} T \cdot \left[\nabla \times \left(\frac{1}{p} \nabla \times U \right) - k_0^2 q U \right] dS = 0, \quad \forall T. \quad (6)$$

This must hold for the exact U and for every admissible T ; it is exactly equivalent to the original PDE holding pointwise, but written in integral form.

2.2 Why weaken it at all

Equation (6) still contains a *second* derivative acting on U (curl of a curl). This demands U be twice-differentiable, which is a strong requirement to impose on simple low-order basis functions. The goal of the next step is to move one curl off of U and onto T , halving the smoothness required of each.

3 Integration by Parts: Vector Identity + Green's Theorem

3.1 The vector identity

For any vector fields $\mathbf{A}, \mathbf{B}$,

$$\nabla \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B}). \quad (7)$$

Set $\mathbf{A} = T$, $\mathbf{B} = \frac{1}{p} \nabla \times U$:

$$\nabla \cdot \left(T \times \frac{1}{p} \nabla \times U \right) = \left(\frac{1}{p} \nabla \times U \right) \cdot (\nabla \times T) - T \cdot \left(\nabla \times \frac{1}{p} \nabla \times U \right).$$

Rearranged to isolate the term we want to replace:

$$T \cdot \left(\nabla \times \frac{1}{p} \nabla \times U \right) = (\nabla \times T) \cdot \left(\frac{1}{p} \nabla \times U \right) - \nabla \cdot \left(T \times \frac{1}{p} \nabla \times U \right). \quad (8)$$

3.2 Integrate and apply the divergence (Green's) theorem

Integrate (8) over Ω :

$$\iint_{\Omega} T \cdot \left(\nabla \times \frac{1}{p} \nabla \times U \right) dS = \iint_{\Omega} (\nabla \times T) \cdot \left(\frac{1}{p} \nabla \times U \right) dS - \iint_{\Omega} \nabla \cdot \left(T \times \frac{1}{p} \nabla \times U \right) dS.$$

Apply the divergence theorem, $\iint_{\Omega} \nabla \cdot \mathbf{F} dS = \oint_{\Lambda} \mathbf{F} \cdot \hat{n} dl$, to the last term with $\mathbf{F} = T \times \frac{1}{p} \nabla \times U$:

$$\iint_{\Omega} \nabla \cdot \left(T \times \frac{1}{p} \nabla \times U \right) dS = \oint_{\Lambda} \left(T \times \frac{1}{p} \nabla \times U \right) \cdot \hat{n} dl. \quad (9)$$

3.3 Assembled weak form

Substituting (8)–(9) back into the $k_0^2 q T \cdot U$ -augmented weighted residual (6):

$$\boxed{\iint_{\Omega} \left[(\nabla \times T) \cdot \left(\frac{1}{p} \nabla \times U \right) - k_0^2 q T \cdot U \right] dS = \oint_{\Lambda} T \times \left(\frac{1}{p} \nabla \times U \right) \cdot \hat{n} dl} \quad (10)$$

What changed: the left side is now symmetric in T and U , with only *first* derivatives of each — exactly the regularity that first-order (Whitney) basis functions provide. The boundary term is not an extra choice; it falls out automatically of the derivation, and it is precisely where absorbing boundary conditions and the incident excitation will enter (Section 6).

4 Galerkin Discretization

4.1 Finite-dimensional approximation

Represent the unknown field by a finite sum of basis (shape) functions N_i :

$$U^h(\mathbf{r}) = \sum_{i=1}^N u_i N_i(\mathbf{r}), \quad (11)$$

with u_i unknown coefficients and N the total number of degrees of freedom.

4.2 Galerkin's choice of test function

Galerkin's method chooses the test functions from the *same* space as the trial functions:

$$T = N_j, \quad j = 1, \dots, N.$$

This is a projection statement: it forces the residual of U^h to be orthogonal to every basis function used to build U^h itself,

$$\iint_{\Omega} N_j \cdot R \, dS = 0, \quad j = 1, \dots, N,$$

which is the closest U^h can get, within the chosen finite-dimensional space, to satisfying the PDE exactly (equivalent, for this self-adjoint operator, to the Rayleigh–Ritz variational minimization).

4.3 Substitution into the weak form

Substitute (11) and $T = N_j$ into (10):

$$\iint_{\Omega} \left[(\nabla \times N_j) \cdot \left(\frac{1}{p} \nabla \times \sum_i u_i N_i \right) - k_0^2 q N_j \cdot \sum_i u_i N_i \right] dS = \oint_{\Lambda} N_j \times \left(\frac{1}{p} \nabla \times U \right) \cdot \hat{n} \, dl.$$

By linearity, pull the sum and the unknown coefficients u_i outside the integral:

$$\sum_{i=1}^N u_i \iint_{\Omega} \left[(\nabla \times N_j) \cdot \left(\frac{1}{p} \nabla \times N_i \right) - k_0^2 q N_j \cdot N_i \right] dS = \oint_{\Lambda} N_j \times \left(\frac{1}{p} \nabla \times U \right) \cdot \hat{n} \, dl. \quad (12)$$

This is one linear algebraic equation for each $j = 1, \dots, N$: exactly N equations in N unknowns u_i . In matrix form,

$$\boxed{A\mathbf{u} = \mathbf{b}}, \quad A_{ji} = \iint_{\Omega} \left[(\nabla \times N_j) \cdot \left(\frac{1}{p} \nabla \times N_i \right) - k_0^2 q N_j \cdot N_i \right] dS, \quad (13)$$

with $\mathbf{b}$ (for now) representing the boundary integral on the right — still to be made concrete in Section 6.

5 Element Matrices and Global Assembly

5.1 Splitting the domain integral into element contributions

The computational domain is tessellated into triangles, $\Omega = \bigcup_{e=1}^{N_e} \Omega_e$. Since integration is additive over disjoint subdomains,

$$\iint_{\Omega} (\cdot) \, dS = \sum_{e=1}^{N_e} \iint_{\Omega_e} (\cdot) \, dS.$$

Define the *element matrix* entries as the same integrand restricted to one triangle:

$$A_{ij}^{(e)} = \iint_{\Omega_e} \left[(\nabla \times N_j) \cdot \left(\frac{1}{p} \nabla \times N_i \right) - k_0^2 q N_j \cdot N_i \right] dS + B_{ij}^{(e)}, \quad (14)$$

where $B_{ij}^{(e)}$ collects the boundary-term contribution associated with element e (derived explicitly in Section 7). On a given triangle, N_i, N_j are nonzero for only the handful of basis functions supported on that element (three edges, for first-order Whitney elements), so $A^{(e)}$ is a small, dense local matrix.

5.2 Assembly into the global system

Because neighbouring elements share edges/nodes, the same global degree of freedom receives contributions from every element touching it. The global matrix is the sum of local matrices scattered into global indices:

$$A = \sum_{e=1}^{N_e} A^{(e)}, \quad A\mathbf{u} = \mathbf{b}. \quad (15)$$

5.3 Why the assembled matrix is sparse

Basis function N_i is nonzero only on the small cluster of elements touching degree of freedom i . Hence

$$A_{ij} = \sum_e A_{ij}^{(e)} \neq 0 \quad \text{only if } i, j \text{ share at least one element.}$$

For degrees of freedom far apart in the mesh, $A_{ij} = 0$ identically. This locality is what allows $A\mathbf{u} = \mathbf{b}$, even with $\sim 10^6$ unknowns, to be solved efficiently with a sparse direct solver.

6 Boundary Conditions: The Bounded-Domain Problem

6.1 Why a boundary condition is needed at all

The physical problem is defined on an unbounded region (semi-infinite soil, open air above). The computational domain $\Omega = \Omega_{\text{air}} \cup \Omega_{\text{soil}}$ must be truncated to a finite region bounded by an artificial contour $\Lambda = \Gamma_a \cup \Gamma_s$. Without special treatment, waves reaching Λ would reflect back into Ω as if off a hard wall, contaminating the solution. An *absorbing boundary condition* (ABC) is used to approximate a purely outgoing wave at Λ , letting energy leave with minimal spurious reflection.

6.2 Incident/scattered field decomposition

Write the total field as

$$U = U_i + U_s, \quad (16)$$

where U_i is the known incident excitation (the Thorsos-tapered plane wave) and U_s is the unknown scattered field. This decomposition is used *only where it is physically valid*:

- **On Γ_a (outer air boundary):** the total field still contains the incoming incident wave, which is not outgoing, so the ABC cannot be applied to U directly. It *can* be applied to U_s , since the scattered field is, by construction, purely outgoing. Hence (16) is used here, and the ABC is enforced on U_s .
- **On Γ_s (outer soil boundary):** U_i is not a valid solution of Maxwell's equations inside a lossy dielectric, so no meaningful incident/scattered split exists in the soil. The total field itself is assumed (approximately) purely outgoing at depth, and the ABC is applied directly to U , with no decomposition.

6.3 The first-order ABC

The first-order absorbing boundary condition, enforcing no reflection of a plane wave normally incident on a planar boundary, reads

$$(\nabla \times U_s) \times \hat{n} = jk_0 \cos \theta (\hat{n} \times (\hat{n} \times U_s)) \quad (17)$$

on Γ_a (with $U_s \rightarrow U$ and the analogous form applied on Γ_s), where θ is the angle between the incident wave vector and the boundary normal $\hat{n}$.

7 Incorporating the ABC into the Weak Form

7.1 Splitting the boundary integral

Substitute (16) into the boundary integral of (10) restricted to Γ_a :

$$\oint_{\Gamma_a} T \times \left(\frac{1}{p} \nabla \times U \right) \cdot \hat{n} dl = \underbrace{\oint_{\Gamma_a} T \times \left(\frac{1}{p} \nabla \times U_i \right) \cdot \hat{n} dl}_{\text{known — depends only on } U_i} + \underbrace{\oint_{\Gamma_a} T \times \left(\frac{1}{p} \nabla \times U_s \right) \cdot \hat{n} dl}_{\text{unknown — depends on } U_s}. \quad (18)$$

The first piece involves only the prescribed incident field: no unknown coefficients u_i appear in it. The second piece still contains an unknown U_s ; it will be rewritten, not eliminated, using the ABC.

7.2 Substituting the ABC

Using (17), $\nabla \times U_s$ can be replaced everywhere it appears at the boundary by an expression involving U_s itself, with no derivative:

$$\left(\frac{1}{p} \nabla \times U_s \right) \times \hat{n} = \frac{1}{p} j k_0 \cos \theta (\hat{n} \times (\hat{n} \times U_s)).$$

After the appropriate vector-triple-product rearrangement (to express the term as a $T \cdot (\dots)$ contraction), the second integral in (18) becomes a purely bilinear expression in T and U_s :

$$\oint_{\Gamma_a} T \times \left(\frac{1}{p} \nabla \times U_s \right) \cdot \hat{n} dl \rightarrow \frac{j k_0 \cos \theta}{p} \oint_{\Gamma_a} T \cdot (\hat{n} \times (\hat{n} \times U_s)) dl. \quad (19)$$

Crucially, since U_s is expanded in the same basis, $U_s^h = \sum_i u_i N_i$, this term multiplies the unknown coefficients u_i — exactly like the volume integral does. An analogous substitution (with $U_s \rightarrow U$, no split) is made on Γ_s .

7.3 Moving the unknown-dependent boundary term to the left side

Because (19) multiplies the unknowns, it does not belong with the genuinely known right-hand side; it must be moved to the same side of the equation as every other term multiplying u_i :

$$\iint_{\Omega} [\dots] dS - \frac{j k_0 \cos \theta}{p} \oint_{\Lambda} T \cdot (\hat{n} \times (\hat{n} \times U_s)) dl = \oint_{\Gamma_a} T \times \left(\frac{1}{p} \nabla \times U_i \right) \cdot \hat{n} dl. \quad (20)$$

Everything on the left multiplies u_i and is discretized exactly like a volume element integral; everything on the right depends only on the known U_i .

7.4 Discretized boundary matrix $B_{ij}^{(e)}$

Setting $T = N_j$, $U_s^h = \sum_i u_i N_i$ in (20), the per-element ABC contribution is

$$B_{ij}^{(e)} = \begin{cases} -\frac{j k_0 \cos \theta}{p} \int_{\Lambda \cap \partial \Omega_e} N_j \cdot (\hat{n} \times (\hat{n} \times N_i)) dl, & \text{if } \Omega_e \text{ has an edge on } \Lambda, \\ 0, & \text{otherwise.} \end{cases} \quad (21)$$

This is nonzero only for the thin ring of elements touching the artificial outer mesh boundary $\Lambda = \Gamma_a \cup \Gamma_s$; it vanishes for every interior element. Adding it into $A^{(e)}$, as in (14), gives

$$A = A_{\text{volume}} + A_{\text{boundary}}, \quad A_{\text{boundary}} = \sum_e B^{(e)}. \quad (22)$$

7.5 The right-hand side

The genuinely known term — the right side of (20) — becomes, for each test function N_j ,

$$b_j = \oint_{\Gamma_a} N_j \times \left(\frac{1}{p} \nabla \times U_i \right) \cdot \hat{n} \, dl, \quad (23)$$

computed entirely from the prescribed incident field U_i (e.g. the Thorsos-tapered excitation), nonzero only on elements touching Γ_a . This is precisely how the incident radar wave enters the FEM system.

8 The Assembled Linear System

Collecting every stage above, the final finite-element system is

$$\boxed{A\mathbf{u} = \mathbf{b}}, \quad A_{ji} = \underbrace{\iint_{\Omega} \left[(\nabla \times N_j) \cdot \left(\frac{1}{p} \nabla \times N_i \right) - k_0^2 q N_j \cdot N_i \right] dS}_{\text{volume term (all elements)}} + \underbrace{B_{ji}}_{\text{ABC term (boundary elements only)}}, \quad (24)$$

$$b_j = \oint_{\Gamma_a} N_j \times \left(\frac{1}{p} \nabla \times U_i \right) \cdot \hat{n} \, dl \quad (\text{known incident-field forcing}).$$

Once solved for $\mathbf{u} = [u_1, \dots, u_N]^T$, the approximate total field is reconstructed as

$$U^h(\mathbf{r}) = \sum_{i=1}^N u_i N_i(\mathbf{r}),$$

from which the scattered field $U_s^h = U^h - U_i$ is extracted, transformed to the far field, and used to compute the radar cross section.

Summary of the derivation chain

Maxwell $\rightarrow$ vector wave eq. $\rightarrow$ weighted residual $\rightarrow$ vector identity + Green's thm.
 $\rightarrow$ weak form $\rightarrow$ Galerkin substitution $\rightarrow$ element matrices
 $\rightarrow$ global assembly $\rightarrow$ incident/scattered split & ABC $\rightarrow A\mathbf{u} = \mathbf{b}$.